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Description

Intelligent transport systems – Management of electronic traffic regulations (METR) – Part 5:
Traffic regulation order (TRO) perspectives

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WD stage

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Foreword

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This document was prepared by Technical Committee ISO/TC 204 Intelligent Transport Systems, Working group 19 Mobility integration.

A list of all parts in the ISO 24315 series can be found on the ISO website.

Any feedback or questions on this document should be directed to the user's national standards body. A complete listing of these bodies can be found at www.iso.org/members.html.

Introduction

The Management of Electronic Traffic Regulations (METR) document series is intended to provide users of road transport infrastructure access to geo-specific, trustworthy, timely, authoritative, machine-interpretable, traffic and transport related rules enacted by jurisdictional entities. This is conceptually and simplified shown in Figure 0.1.

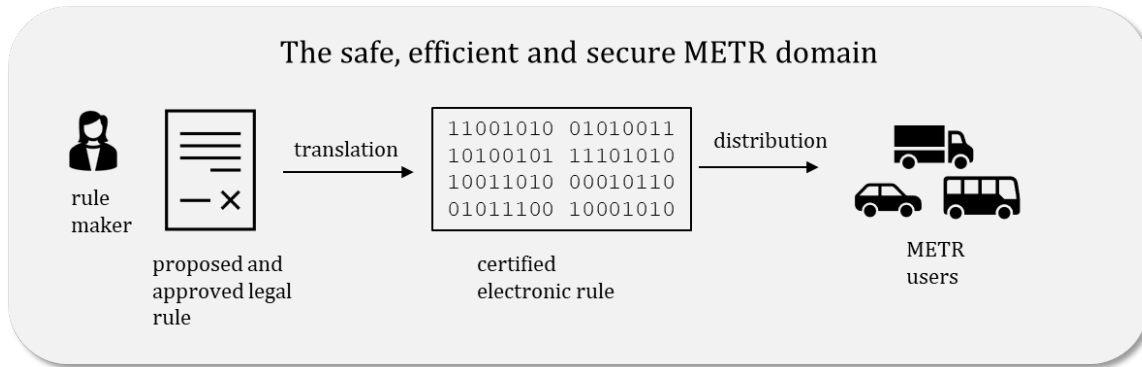


Figure 0.1 The conceptual view of METR

The ISO 24315 METR series of documents is being developed to cover a worldwide implementation of the METR concept. This implies a generalisation of the concept to provide a generic platform for regional and national implementations. A generic platform defined in a series of standards, specifications, and technical reports is a safe, stable, interoperable, and efficient starting point for such regional and national implementations.

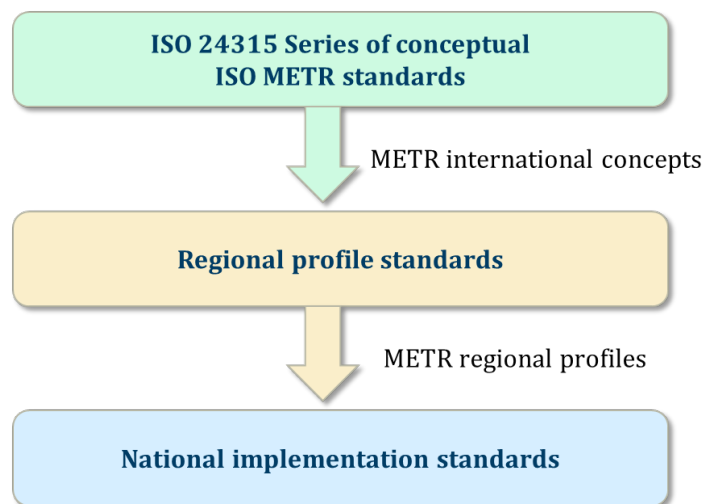


Figure 0.2 — The METR hierarchy of international, regional, and national METR standards

METR is designed to assist developers and manufacturers of driving automation systems (i.e., automation Levels 1-5) and driver information systems (including those at automation Level 0) to electronically obtain traffic rules to better enable:

- interacting safely with other road users

- following instructions from law enforcement organizations, and those authorized to direct traffic
- maintaining smooth and safe flow of traffic; and
- complying with other rules enacted to support legislative policies (such as environmental protection, noise, manage height and weight restrictions, and societal aspects such as market days, fiestas, pedestrian zones, etc.).

A Traffic Regulation Order (TRO) is a legal document that defines one or more rules. The legal contents of a TRO can vary by jurisdiction, e.g. region, country, state, city, as directed by overarching legislation within the jurisdiction. This is beyond the scope of this standard.

Rules includes rules of the road, regulations, advisories and guidance. The term Rules of the road is defined as component of the highway code that specifies foundational rules for the operation of vehicles. However, highway code, aka as road code, can also provide rules concerning non-vehicle users of the highway/road. Rules of the road are not considered relevant for TROs.

Terms and definitions used in this document are based on ISO 24315 METR - Part 1 Vocabulary. However, terms and definitions used in regional, national and local legislation can vary from one region to another, from one nation to another and from one local, e.g. state, county, city or jurisdictional area to another.

Intelligent transport systems – Management of electronic traffic regulations (METR) – Part 5: Traffic regulation order (TRO) perspectives

1 Scope

This document describes the main perspectives related to Traffic Regulation Order (TRO) (aka rule order) which is defined in ISO 24315-1 as a ‘legally recognised document that officially specifies one or more rules’ [SOURCE: ISO 24315-1, 3.2.1.14]. A rule is defined as ‘information regarding allowed or recommended behaviour as established by a rule maker for its jurisdictional area’ [SOURCE: ISO 24315-1, 3.2.1.13].

The perspectives described in this document covers the following subjects:

- Traffic regulation order (TRO), definition and relationship with traffic rules
- jurisdictional entity role in relation to TRO
- rule states
- TRO states
- rule and TRO representations
- coordination of the activation of rules in the physical and digital infrastructure

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

- ISO/TS 14812, *Intelligent transport systems — Vocabulary*
- ISO/TS 24315-1, *Intelligent transport systems — Management of electronic traffic regulations (METR) – Part 1: Vocabulary*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO 14812, and ISO 24315-1 apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- ISO Online browsing platform: available at <https://www.iso.org/obp>
- IEC Electropedia: available at <http://www.electropedia.org/>

4 Symbols and abbreviated terms

TRO	Traffic Regulation Order
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5 Traffic Regulation Order (TRO)

A traffic regulation order (aka as rule order) is a legal document that specify one or more rule(s). A TRO forms part of the system of laws, rules, and measures designed to control, guide, and organise road traffic to ensure safety, efficiency, and order.

The TRO can describe one or more rule(s) that can be rules of the road, regulatory, advisory or guiding [SOURCE: ISO 24315-1, 3.2.1.14, Note 3].

NOTE 1: The term Rules of the road is defined as component of the highway code that specifies foundational rules for the operation of vehicles (SOURCE: 24315-1, 3.2.1.16). However, highway code, aka as road code, can also provide rules concerning non-vehicle users of the highway/road. Rules of the road are not considered relevant for TROs.

NOTE 2: Guiding rules are in some jurisdictional areas, e.g. on a national level, not considered as TROs

A TRO can contain multiple rules - each rule has a clear regulatory requirement, a defined location and conditions that apply (and exemptions that are permitted).

NOTE 2: The legal contents of a TRO can vary by jurisdiction, e.g. region, country, state, city, as directed by overarching legislation within the jurisdiction. This is beyond the scope of this document.

NOTE 3: A TRO contains certain core elements of information and can be extended to support additional parameters that can be defined by jurisdiction.

Figure 1 shows some typical traffic signs related to the different types of rules. Other regions and nations can deploy other traffic signs with the same meaning.

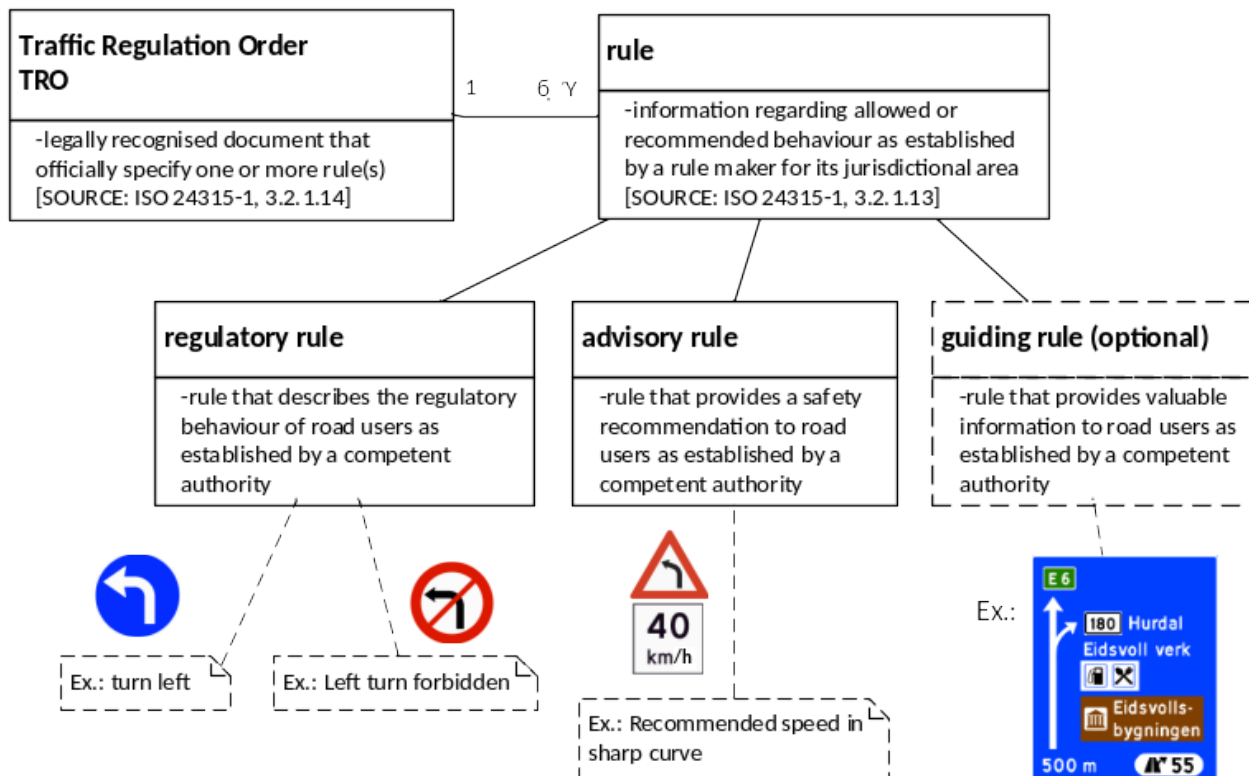


Figure 1 TRO – rule relation and rule examples

The TRO, that can consist of text, drawings, figures, maps with information, pictures etc., is established by a rule maker or competent authority. The rule maker (say local authority) can only define TROs within the overarching constraints and legislation they are allowed to use (as typically defined by national government).

EXAMPLE 1: A legal document including a plan for parking restrictions in an urban area, e.g. part of a city centre or a residential area, showing all required traffic signs and road markings to meet the parking restrictions requirements. The TRO can in this case define multiple parking spaces along multiple streets and different time restrictions.

EXAMPLE 2: A legal document including a plan for a reduced speed limit, e.g., on a road passing a school or in a road network in a residential area, where the TRO shows the speed limits signs on a map including coordinates for each speed limit sign and possibly any time-based conditions. A simplified example is shown in Figure 2.

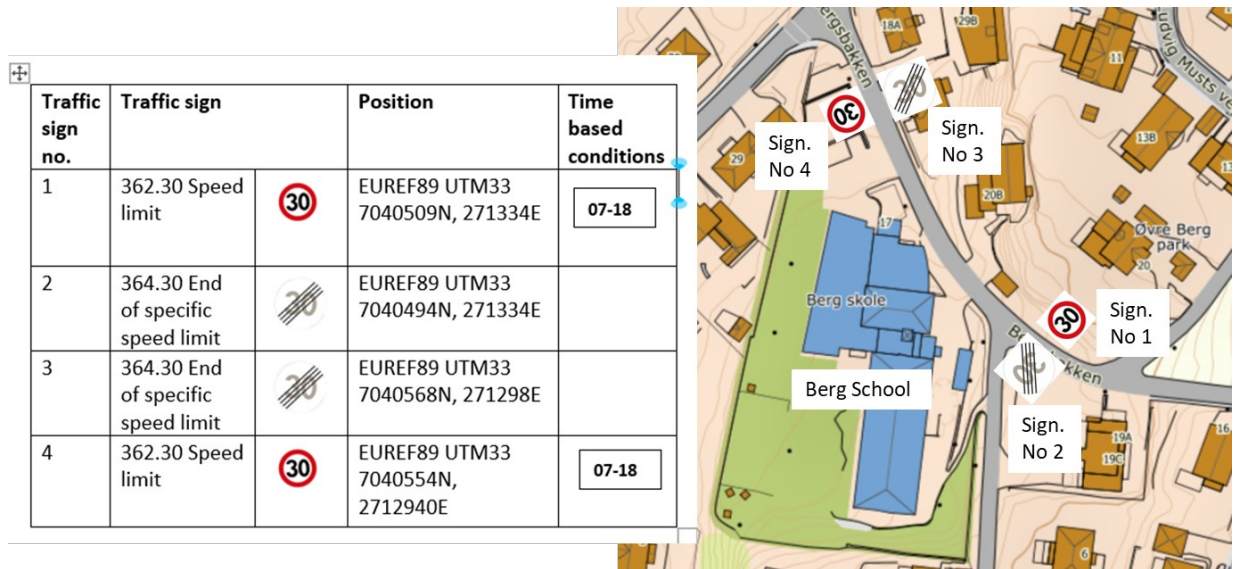


Figure 2 A simplified example on a part of a TRO

EXAMPLE 3: A legal document including a plan for a work zone showing all the traffic control devices (TCD) needed for the protection of the road workers and an effective and safe movement of vehicles, pedestrians and cyclists passing through the work zone. The TRO can also include transition plans defining the order in which rules are deployed, e.g., first posting lower speed limits followed by deployment of traffic cones for motor vehicles, followed by traffic cones for micromobility, etc. While the TRO explains the final state, it also can accommodate a transition period of a process to deploy equipment.

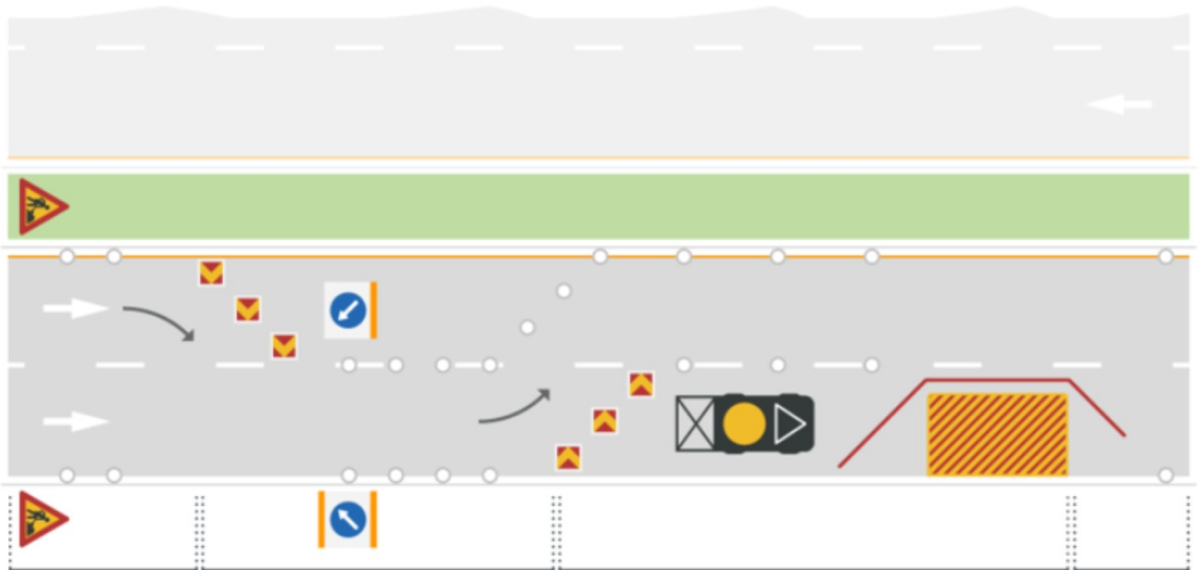


Figure 3 Part of a TRO for a work zone on a motorway (Source: [2])

EXAMPLE 4: A digital TRO form filled in and digitally signed by a rule maker, e.g. authorised personnel at a Traffic Management Centre (TMC), in case of a road accident and an urgent need for a temporary speed limit reduction and closing of lanes. The incident could also be handled by activating a dormant TRO.

6 Jurisdictional entity

A jurisdictional entity (aka jurisdictional authority) has the legal power to issue rules of behaviour for road infrastructure users. Each jurisdictional entity has an associated jurisdictional area for which the jurisdictional

entity has the power to define rules of behaviour and these areas can overlap. For example, any specific location can be subject to multi-national, national, sub-national, and local jurisdictions. In fact, activities on private property are also potentially subject to rules defined by the landowner (e.g., a large university imposing rules of campus streets).

NOTE: Legal implications related to defining rules for private roads and the enforcement of those rules vary by parent jurisdiction (e.g., national and local policies).

The authority to establish rules at a particular location can depend upon several factors, including type of facility, type of rule, etc. For example, a city can be responsible for defining most rules for most transport facilities within its jurisdictional area, while the county defines the most rules for county roads and the national government defines most rules for motorways. However, even in this case, the city could still be granted the authority to issue temporary rules on the motorway (e.g., if the city police are responsible for responding to real-time conditions). In general, it is the responsibility of the higher-level jurisdictional entities to establish the policy framework that defines who has the authority to establish which type of rules.

A rule maker has the responsibility for creating and maintaining rules for a geographic and operational scope as defined by the parent jurisdictional entity. Within each jurisdictional entity there can be multiple rule makers with different scopes, such as:

- a legislative entity (e.g., city council) that defines rules that are subject to deliberative political processes (e.g., default speed limit on city streets)
- an administrative entity that defines rules based on engineering principles (e.g., speed limit for specific segment of roadway)
- an operations entity (e.g., police or traffic management centre) that defines or activates rules in response to real-time conditions (e.g., lane closures, emergency response plans)

The framework for directing the administrative responsibilities of each jurisdictional entity and rule maker are often defined by the higher-level jurisdictional entities. For example, the national jurisdictional entity can define a law that requires each city jurisdictional entity to define certain rules. In response, the city jurisdictional entity can delegate a portion of its responsibilities to an administrative or operations entity.

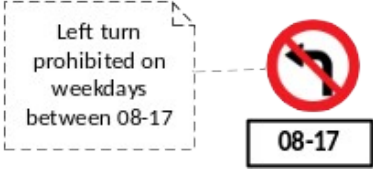
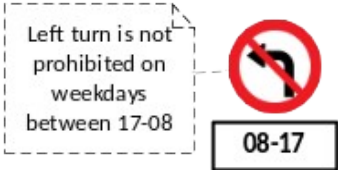
7 Rule states

A rule will have different states, e.g., a rule can be proposed or implemented. Table 1 shows the different rule states and some examples on the terms active and inactive rules.

NOTE: This table extends and revises the states defined in ISO 24315-1:2025. The documents are expected to be harmonized in a future edition.

Table 1 — Rule states

Rule state	Definitions	Comment
<i>active</i>	rule that is implemented and presently in effect [SOURCE: ISO 24315-1:2025, 3.2.4.1]	A rule can only be enforced when active and a rule can be continually active during certain hours of the day or upon other defined conditions [SOURCE: ISO 24315-1:2025, 3.2.4.1, NOTE 1 and 2]. This typically requires both legal TRO to be in force and appropriate indicators to be present on the road (signs and markings). In the case of posted rules, appropriate TCDs must be implemented (i.e., installed and enabled) prior to enforcement. EXAMPLE:

Rule state	Definitions	Comment
		
<i>annulled</i>	rule that has been declared invalid, void, or unenforceable by a court or other competent authority, and as a result, any prior enforcement or application of such rules shall cease to have effect.	An annulled rule is a rule that is completely withdrawn as if it never existed, often because it was improperly made, erroneous, or unlawful.
<i>approved</i>	rule that has been approved by a rule maker with a future inception time [SOURCE: ISO 24315-1:2025, 3.2.4.2]	Inception time is the point in time at which a rule can first become applicable whenever all defined conditions are met [SOURCE: ISO 24315-1, 3.2.4.5]. It could be approved but not implemented - the inception time (for road indicators) could be in the past.
<i>beingImplemented</i>	rule that is being implemented	After approval, a posted rule requires time to install and authorize associated TCDs. The beingImplemented state represents this period.
<i>dormant</i>	rule that has been approved by a rule maker but requires additional legal action in engage	Activating any dormant rule requires a rule maker to formally announce the engagement of that rule. A dormant rule is a rule that can very seldom or never be implemented. It could, e.g., be a rule used for specific emergency cases like evacuation of citizens, avalanches, landslides and earthquakes.
<i>implemented</i>	rule that has an inception time in the past and a termination time either undefined or in the future [SOURCE: ISO 24315-1:2025, 3.2.4.3]	Inception time is the point in time at which a rule can first become applicable whenever all defined conditions are met [SOURCE: ISO 24315-1, 3.2.4.5]. Termination time is the point in time at which a rule can no longer be enforced [SOURCE: ISO 24315-1, 3.2.4.10]. An implemented rule can be active, inactive or overridden [SOURCE: ISO 24315-1, 3.2.4.3 NOTE 1]. For rules based on certain conditions of operation a rule could be implemented but not currently active.
<i>inactive</i>	rule that is implemented but not presently in effect [SOURCE: ISO 24315-1:2025, 3.2.4.4]	A rule can be restricted based on conditions, such as time-of-day. The rule cannot be enforced while it is inactive [SOURCE: ISO 24315-1, 3.2.4.4, Notes 1 and 2]. EXAMPLE: 

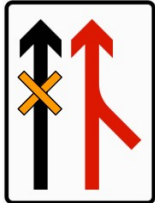
Rule state	Definitions	Comment
<i>overridden/suspended</i>	rule that is implemented but temporarily nullified or replaced by another rule [SOURCE: ISO 24315-1:2025, 3.2.4.6]	It is expected that the rule that overrides another rule will eventually terminate [SOURCE: ISO 24315-1, Note 1], even if the expected termination time is unknown. EXAMPLE:  Overriding rule
<i>proposed</i>	rule that is not yet approved by the rule maker [SOURCE: ISO 24315-1:2025, 3.2.4.7]	
<i>terminated</i>	rule that was previously in effect	A rule can be terminated by its termination time, formally cancelled or annulled by an authority or through a new rule that replaces or nullifies it.

Figure 4 shows a rule state diagram based on the rule states defined in Table 1.

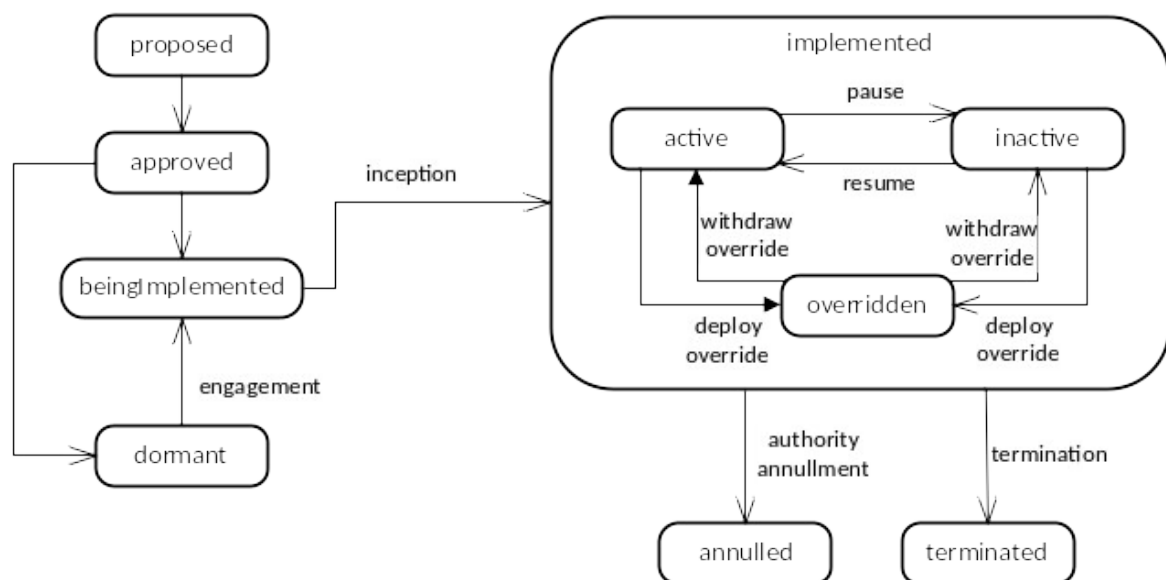


Figure 4 Rule state diagram

8 TRO states

A TRO will be in different states depending on the position in the life cycle of the TRO, e.g., a TRO can be proposed or annulled. Table 2 — TRO states describes the different TRO states.

NOTE: This table extends and revises the states defined in ISO 24315-1:2025. The documents are expected to be harmonized in a future edition.

Table 2 — TRO states

TRO state	Definition	Comment
<i>annulled</i>	TRO that has been declared invalid, void, or unenforceable by a court or other competent authority, and as a result, any prior enforcement or application of the rules within the TRO shall cease to have effect.	An annulled TRO is a TRO that is completely withdrawn as if it never existed, often because it was improperly made, erroneous, or unlawful.
<i>approved</i>	TRO that has been approved by a rule maker with a future inception time for all rules within the TRO	Inception time is the point in time at which a rule can first become applicable whenever all defined conditions are met [SOURCE: ISO 24315-1, 3.2.4.5].
<i>dormant</i>	TRO that has been approved by a rule maker but none of the rules within the TRO are engaged	A dormant TRO may very seldom or never be engaged, e.g., a TRO used for emergency preparedness and evacuation of citizens.
<i>in operation</i>	TRO that has come into force and is currently effective, such that its rule(s) are applicable and enforceable within the relevant jurisdiction at the present time.	The state does not prevent that the rule(s) within the TRO may individually change between the states active, inactive and overridden.
<i>proposed</i>	TRO that is not yet approved by the rule maker	
<i>terminated</i>	TRO that was previously in effect	A TRO can be terminated by the rule(s) termination time(s), formally cancelled or annulled by an authority or through a new TRO that replaces or nullifies it.

The TRO states are shown in Figure 5.

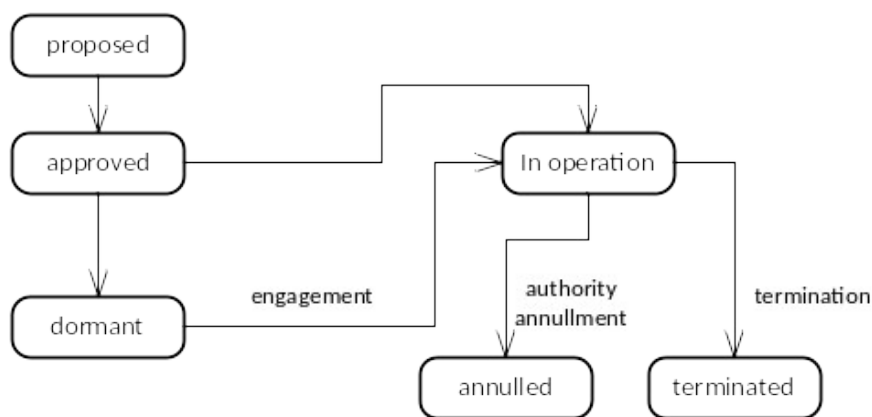


Figure 5 TRO state diagram

9 Rule and TRO representations

A rule can be represented in different formats which is shown in Table 3.

Table 3 — rule representation

rule representation	Definition	Comment
<i>electronic rule</i>	representation of a rule in a structured, digital format that can be readily processed by a computer to determine the intent of the rule [ISO 24315-1, 3.2.6.1]	The structured and digital format is a digital representation of the human-readable text, drawings, figures, maps with information, pictures etc. that defines and describes the rule. An approved electronic rule could be the METR rule.
<i>legal rule</i>	representation of a rule as recorded in official records of the jurisdictional entity [ISO 24315-1, 3.2.6.2]	The legal and human readable rule is the master of a rule. As both the legal and electronic representations will be the basis for enforcement, it is of crucial importance that the two representations are equivalent and that the equivalence is verified and certified. This is why jurisdictional entities are encouraged to adopt an electronic format for their rules.
<i>METR rule</i>	electronic rule that is produced and distributed in a trustworthy manner [ISO 24315-1, 3.2.6.3]	The electronic rule can consist of a set of files in different digital formats. The translation of the different digital formats to secure METR formats, will ensure security and interoperability for the METR users.
<i>Physical rule</i>	representation of a rule as presented in the field [ISO 24315-1, 3.2.6.4]	EXAMPLE: The speed limit as displayed in the field  SOURCE: vegkart.no

A TRO can be represented in different ways depending on its state.

NOTE: This table extends and revises the representations defined in ISO 24315-1:2025. The documents are expected to be harmonized in a future edition.

Table 4 — TRO representation

TRO representation	Definition	Comment
<i>electronic TRO</i>	representation of a TRO in a structured, digital format that can be readily processed by a computer to determine the intent of a TRO and its rules [based on ISO 24315-1, 3.2.6.1]	The structured and digital format is a digital representation of the human-readable text, drawings, figures, maps with information, pictures etc. that defines and describes the rules within the TRO and the TRO specific information. An approved electronic TRO could be the METR TRO.
<i>legal TRO</i>	representation of a TRO as recorded in official records of the jurisdictional entity [based on ISO 24315-1, 3.2.6.2]	The legal and human readable TRO is the master of a TRO. As both the legal and electronic representations will be the basis for enforcement, it is of crucial importance that the two representations are equivalent and that the equivalence is verified and certified. This is why jurisdictional entities are encouraged to adopt an electronic format for their TROs.
<i>METR TRO</i>	electronic TRO that conforms to the METR requirements	The electronic TRO can consist of a set of files in different digital formats. The translation of the different digital formats to secure METR formats, will ensure security and interoperability for the METR users.

10 TRO proposal and approval

Figure 6 shows the processes related to the TRO states Proposed and Approved.

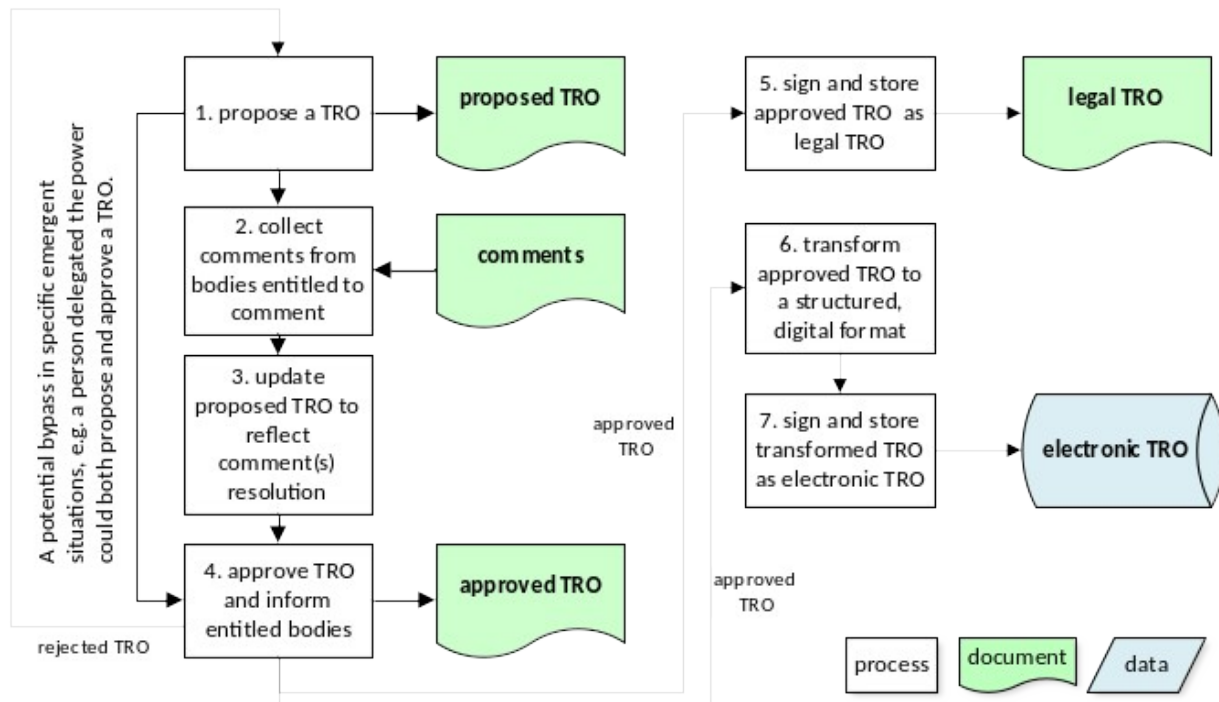


Figure 6 — TRO proposal and approval

A rule maker, appointed by a jurisdictional entity, e.g. a Road Transport Authority (RTA), proposes a TRO and distributes the proposed TRO to bodies entitled to comment on the proposed TRO. Examples on RTAs are a national road administration for national and regional roads and a city council for the road network inside the city borders. Examples on bodies entitled to comment on proposed TRO are emergency services like police, health and fire, local transport authorities and Public Transport Operators (PTO). The bodies entitled to comment on the TRO send their comments back to the rule maker in charge of the TRO. The rule maker updates the TRO based on the accepted comments and prepare the legal document and the included annexes (plans, maps, drawings, pictures, etc.) which then becomes the approved TRO.

The bodies entitled to comment can be informed about the approved TRO. A typical example is a body that has the mandate to enforce the TRO, e.g. the police or a parking agency. For permanent orders this can also include the general public via a consultation.

The TRO can be stored as a legal TRO (document) and as an electronic TRO, i.e. an electronic representation in a structured, digital format that can be readily processed by a computer to determine the intent of the TRO. The legal TRO will be signed by the rule maker signature and possibly stamp, and the electronic TRO will be signed by a digital signature and certificate ensuring the authenticity of the TRO.

The rule maker can be centrally located, e.g. a road administration or a Traffic Management Centre (TMC), but it can also be a person located in the field where the TRO will be implemented. This could, e.g. be a contractor responsible for road works or a person authorised to handle an incident where a TRO is needed to safeguard the travellers, handle vehicles and ensure a safe and effective traffic management. The rule maker could in these cases be mandated to propose and approve the emergent TRO without the comments from bodies entitled to comment on TROs. The centrally located rule maker can also establish emergent TROs, e.g. a Traffic Management Centre (TMC) can support a contractor or a police officer in emergent situations. A rule maker located in the field can also activate a TRO already approved by changing the status from inactive to active.

The delegation of the power to propose and approve a TRO depends on the national organisation of jurisdictional entities and rule makers and is a matter of national legislation and regulations.

11 TRO installation and implementation

An approved TRO will in most cases be installed and implemented. In the TRO context the term ‘installed’ means the physical installation of the measures, e.g. traffic control devices (TCD) that can be described in the approved TRO. The term ‘implemented’ means that the TCD is made active, e.g. traffic signs that have been covered are uncovered. TCD is defined in [1] as ‘physical infrastructure used to visually, audibly or tangibly convey information to travellers’. Examples on TCD are signs, signals, markings and physical barriers.

Some TROs are made for emergency preparedness and can never be installed or implemented (jurisdiction dependant). Some TROs will also be sequentially installed and implemented, e.g., a TRO with parking regulations covering all streets in a small city can be sequentially installed and implemented.

The TRO is installed by the ‘rule installer’ and implemented by the ‘rule implementer’. A rule installer is defined in [1] as ‘role that is responsible for posting rules for a jurisdictional area using conventional traffic control devices’. A rule implementer is defined in [1] as ‘role that is responsible for enabling a rule’.

Figure 7 shows the two flows for the installation and implementation of the approved TRO. The flow to the left is the installation and implementation of the TRO in the physical infrastructure. The flow to the right is the sequence of processes in the METR digital infrastructure leading to the activation of the METR information representing the TRO in an electronic and METR format.

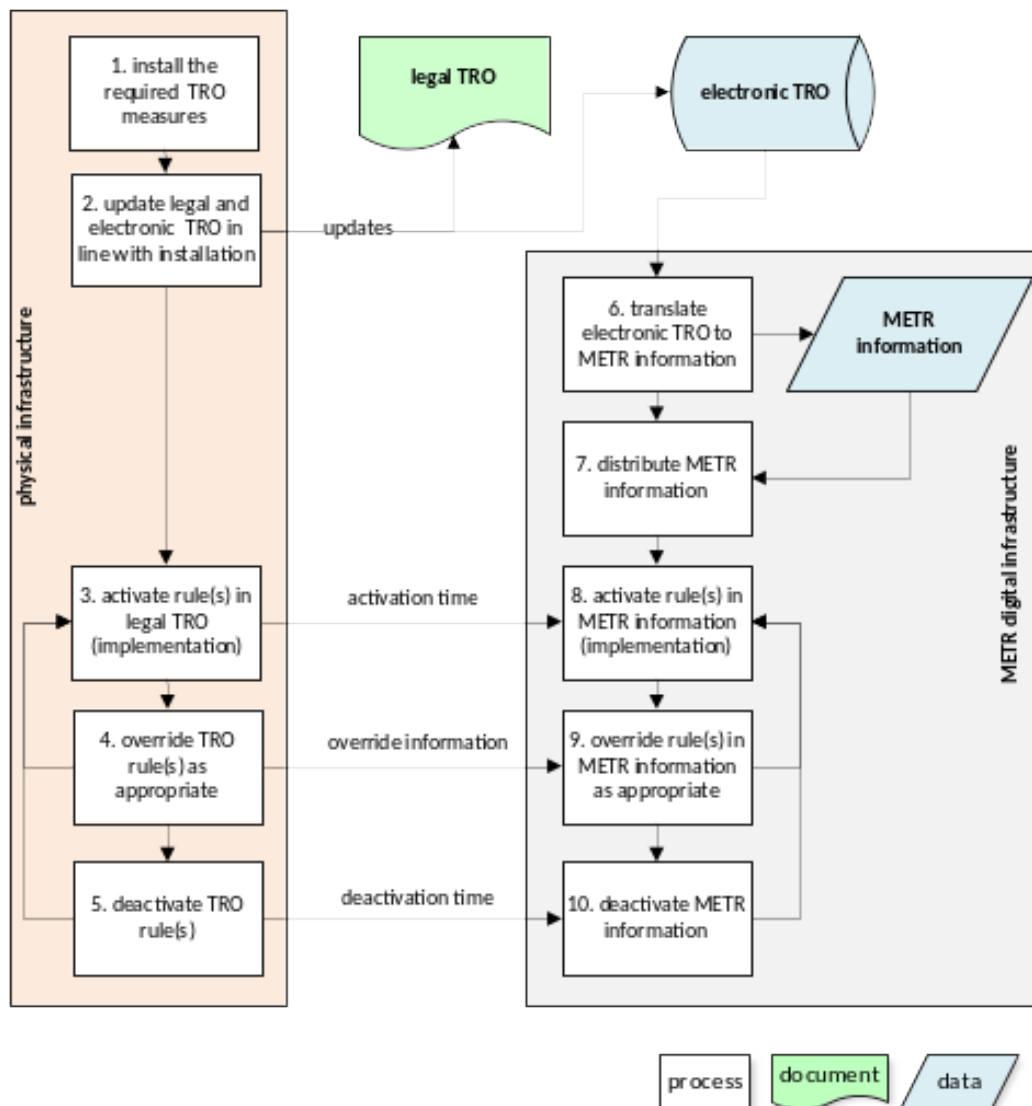


Figure 7 — TRO installation and implementation

The rule installer could make changes in the physical installation due to local conditions and circumstances, e.g., moving a traffic sign to improve the sight to the traffic sign or add another traffic sign to ensure the non-METR user observation and understanding of the traffic sign.

When the rule installer has installed the approved TRO, the rule installer can measure the accurate coordinates of the traffic control devices (TCDs), e.g., traffic signs, stop lines and other markings, traffic signals, barriers etc. and forward the coordinates to the rule maker that updates the TCD coordinates in the approved TRO. An accurate, reliable and trusted position of the TCDs is required for most ITS and C-ITS applications, e.g., applications that need accurate information on lane markings, stop lines and pedestrian crossings. Some examples on TCDs for a pedestrian crossing controlled by traffic signals are shown in Figure 8. In a METR context wrong coordinates for the stop line could have fatal consequences.



Source: Google maps

Figure 8 — Examples on TCDs in a TRO for a pedestrian crossing

The TRO as represented in the physical infrastructure will be used by both non-METR users and METR users. The first category will be able to observe, understand and follow the traffic signs, traffic signals, markings, barriers etc. that together represents the TRO. For the METR users, the Approved TRO will be translated and verified to METR information distributed in the METR digital infrastructure. The translation will represent the Approved TRO in a METR accepted format enabling an automated vehicle (AV) with an Automated Driving System (ADS) to perform the Dynamic Driving Tasks (DDT) fulfilling the approved TRO requirements in the same way as a human driver. While the human driver (non-METR user) will observe the traffic control devices (TCD) defined in the approved TRO, the METR information representing the TRO must be distributed by different means to the METR user. Typical examples are distribution from a central system by mobile network or by roadside system by short range communication like ITS-DSRC. In addition to the METR information the METR

user can also benefit from external information collected by vehicle sensors, e.g., cameras recognising traffic signs and markings.

12 Coordination of the activation of rules in the physical and digital infrastructure

Each rule has a start (inception) time, which is the time when the rule is implemented for the non-METR users that are depending on observations of TCDs. At the same time, it will preferably be implemented in a METR format for METR users. A TRO may include several rules with different inception times. Figure 9 shows that one TRO may consist of 1 to many rules with individual inception and termination times. The rule attributes in the figure are just examples of possible data. The actual attributes defined for a METR rule are defined in a separate part of the ISO 24315 series and can be further extended with regional or local customizations. Examples on assumed rule attributes can be *orderDurationType* (permanent, temporary), *geographicValidity* (GIS coordinates or linear references), *applicableRoadUsers* (e.g. heavy goods vehicles), *recommendedTCDs* (traffic signs and markings TCD location data).

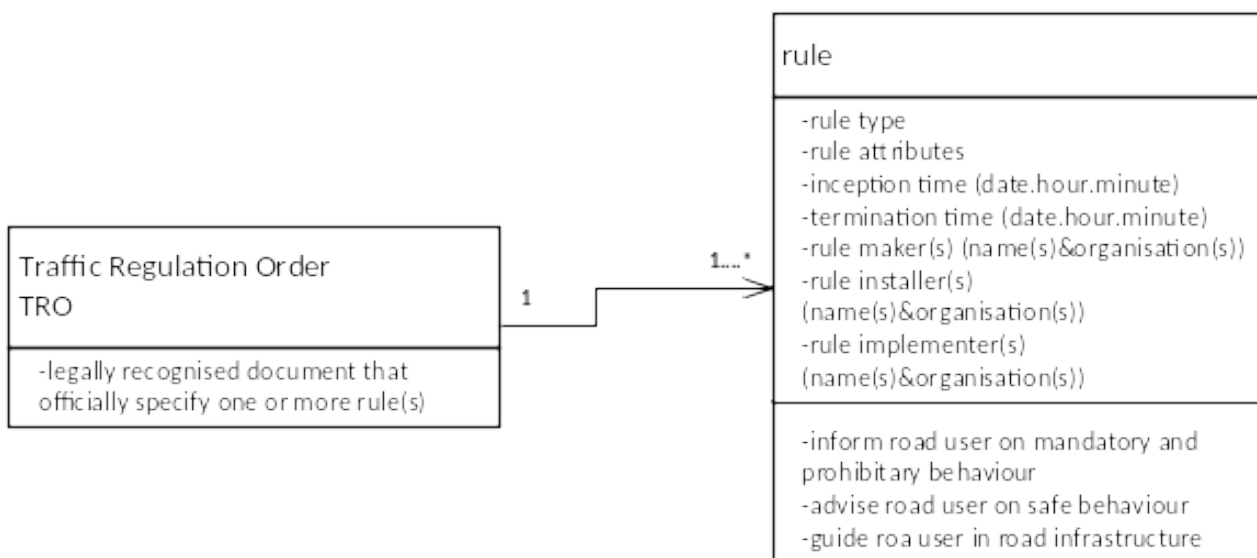


Figure 9 Examples on data defining a rule

The METR information defining a TRO can be distributed before the inception of the rules in the TRO as the inception will be part of the METR information distributed from the METR distributor.

Rules that are approved but not yet implemented can have their TCDs hidden before their inception. E.g., variable message signs (VMS) are black or covered and ordinary traffic signs are covered or clearly marked as not active. This implies that the distribution of the METR information in the METR digital infrastructure and the posting of a TRO in the physical infrastructure is synchronised in a way that the approved TRO is available for non-METR users and METR users at the same time in an ideal scenario. This can be designed in a way to minimize the gap between implementing the legal rule and the METR rule.

In a real-world implementation of a TRO, it will be difficult to perfectly synchronise the implementation in the physical infrastructure and the METR digital infrastructure. For example, in the simplified example in Figure 10 where the TRO contains only one rule, there is a need for a grace period where the enforcement of the one rule should be thoroughly considered and carried through. The grace period should be as short as possible as differences in activation times could cause dangerous incidents. The minimum grace period should be defined by the national transport authorities. It will depend on the type of traffic control devices installed and the risks related to the possible delay in the inception times for physical infrastructure and METR digital infrastructure.

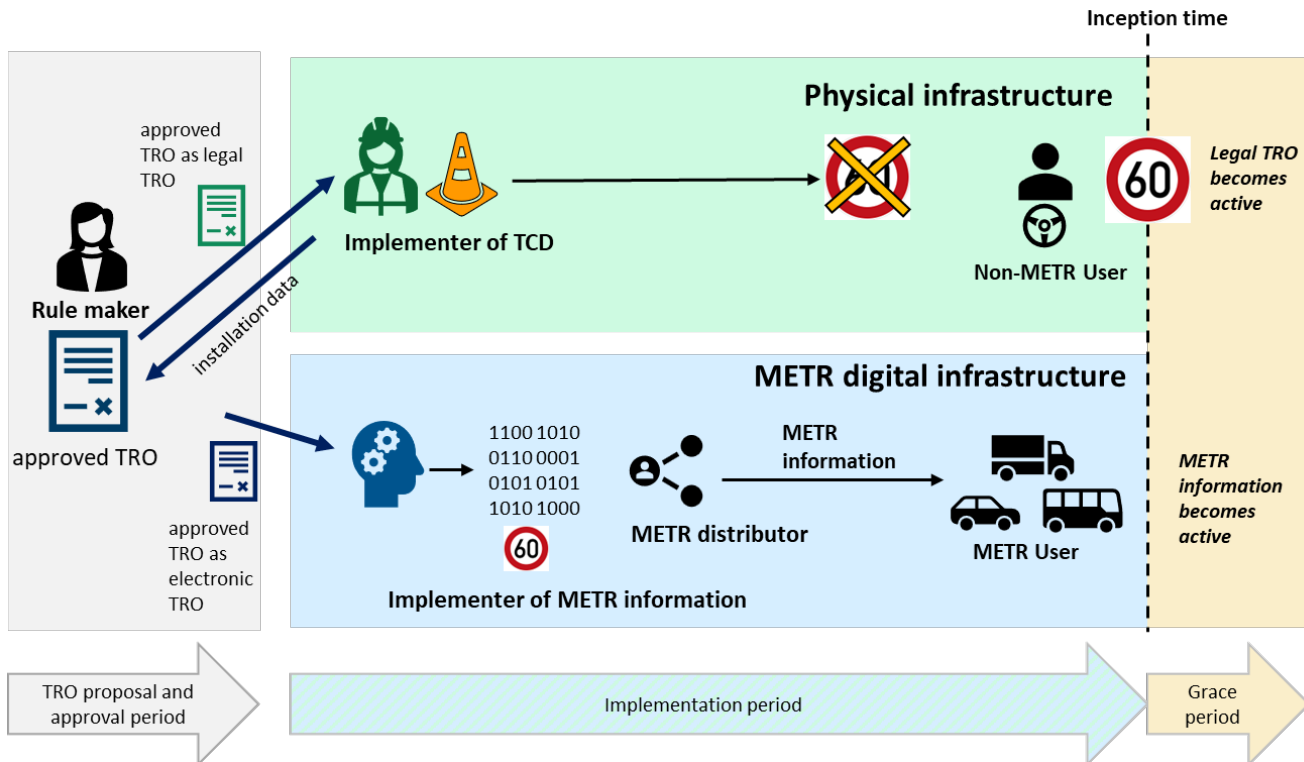


Figure 10 Simplified figure on implementation in a physical and digital infrastructure

The implementer of the TCD can coordinate with the implementer of the METR information to synchronize the implementation of the rule. However, even then there is likely to be some time gap (detectable by a computer) between the implementation of the two representations of the same rule. The human factors related with imposing most new posted rules (even without an electronic version) justify having a grace period for enforcement actions. While the grace period can extend for weeks or months for non-critical regulations in some cases, the grace period can be quite short (e.g., a new stop sign). In these scenarios, it is important that the implementer of the TCD and the implementer of the METR information work in cooperation and can necessitate the integration of the METR regulation system and the maintenance and construction management systems.

An emergent situation and a need for an emergent TRO will typically occur in events like accidents, flooded roads, avalanches and landslides, where a stopped vehicle is blocking a lane, animals on the road, heavy rain, strong wind, fires, dense fog, ice and snow covering lane markings, black ice, and snow- or sandstorm.

The process of defining the TRO in an emergent situation can be slightly different from a non-emergent situation. The emergent situation will most likely require an immediate action from the rule maker that is responsible for the traffic management at the emergent situation location. In many cases there will probably be no time for collecting comments from other bodies entitled to comment. There are two obvious solutions in an emergent situation:

1. The rule maker proposes a new TRO that is installed and translated into a METR information message. An example could be an avalanche that closed a road on a location where an avalanche never previously has closed the road. The TRO would include barriers and traffic signs to be installed.
2. The rule maker activates an existing TRO that already has been installed and translated into a METR information which is earlier distributed but has the state inactive. An example could be a reduced speed limit on a motorway where unexpected black ice occurred. The traffic signs were already installed enabling the rule maker or authorised entity to send a message to the traffic signs to show the reduced speed and at the same time broadcast an activation of an approved or dormant METR information stored by the METR user.

13 Any other perspective to be described

Tbd

14 Any other perspective to be described

Tbd

Bibliography

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Annex A
(informative)

Annex title e.g. Example of a figure and a table

A.1 Clause title

A.1.1 Subclause title

Type text.

A.1.2 Subclause title

A.1.2.1 Subclause title

Type text.

A.1.2.2 Subclause title

Type text.

A.2 Clause title

An example of a figure is given in Figure A.1.

Figure A.1 — Clause title